

Lecture 3

Review of Ch. 1.

A little bit of Chapter 2, and a lot of
Chapter 3: Energy Bands, Electrons and Holes

Streetman and Banerjee, 'Solid State Electronic
Devices,' 6th ed., Pearson, 2006.

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EE 133 Announcements

- Homework 1 is posted on Canvas – Due Sun. 11:59 pm.

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Review Lecture 2 (Ch. 1 material)

- Semiconductors
 - IV: Si, Ge, C
 - III-V: GaAs, InP, InAs, GaP, etc.
- Crystals
 - Lattice:
 - Basis:
 - Primitive cell:
 - Primitive lattice vectors
 - Cubic unit cell:
 - Lattice constant
 - Specific lattices: SC, BCC, FCC
 - Diamond and Zinc Blende crystal structure

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Review Lecture 2 (Ch. 1 material)

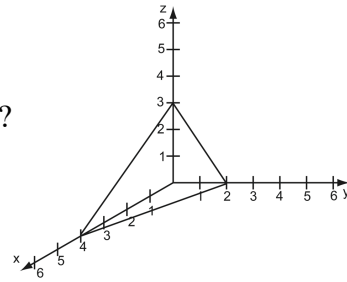
- Crystal directions $[100]$.
- Equivalent directions $\langle 100 \rangle$
- Crystal planes (100) .
 - Miller Indices
 - Find intercepts, invert, multiply by smallest no. to get all integers.
 - For cubic, plane normal to the direction.
- Equivalent planes $\{100\}$
- Si crystal growth technology
 - Czochralski – ingot.
 - Epitaxial: VPE or CVE and MBE.

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Examples

- Crystal planes

What are the Miller indices of plane at right?



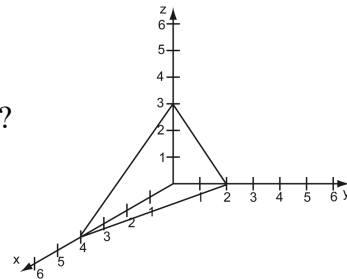
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Examples

- Crystal planes

What are the Miller indices of plane at right?

Intercepts (x, y, z): 4, 2, 3



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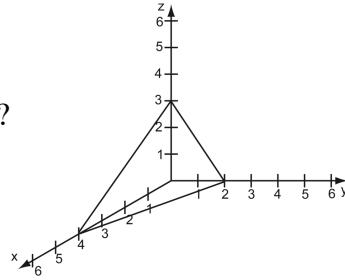
Examples

- Crystal planes

What are the Miller indices of plane at right?

Intercepts (x, y, z): (4, 2, 3)

Inverse of the intercepts ($1/x$, $1/y$, $1/z$): ($1/4$, $1/2$, $1/3$)



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Examples

- Crystal planes

What are the Miller indices of plane at right?

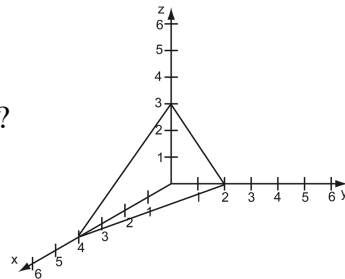
Intercepts (x, y, z): (4, 2, 3)

Inverse of the intercepts ($1/x$, $1/y$, $1/z$): ($1/4$, $1/2$, $1/3$)

Multiply through by lowest common denominator: 12 ($1/4$, $1/2$, $1/3$) =

(3 6 4)

Note: no commas are used to separate Miller indices.



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Examples

- Crystal planes

What are the Miller indices of the x-y plane?

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Examples

- Crystal planes

What are the Miller indices of the x-y plane?

Shift plane away from the origin (0,0,0). Shift it up by 1.

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Examples

- Crystal planes

What are the Miller indices of the x-y plane?

Shift plane away from the origin (0,0,0). Shift it up by 1.

Write down the (x, y, z) intercepts:

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Examples

- Crystal planes

What are the Miller indices of the x-y plane?

Shift plane away from the origin (0,0,0). Shift it up by 1.

Write down the (x, y, z) intercepts: $(\infty, \infty, 1)$

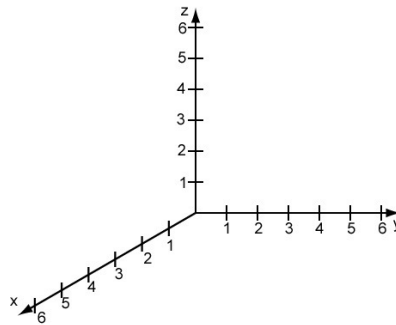
Invert the intercepts $(1/x, 1/y, 1/z)$: $(1/\infty, 1/\infty, 1) = (0\ 0\ 1)$

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Examples

- Crystal planes

Draw the (142) plane.



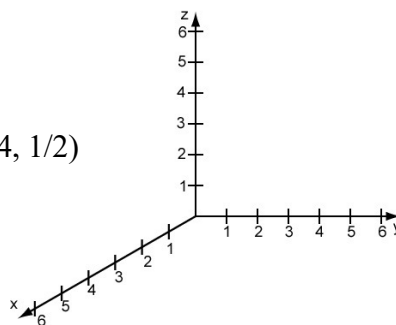
13

Examples

- Crystal planes

Draw the (142) plane.

Invert the Miller indices: $(1, 1/4, 1/2)$



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Examples

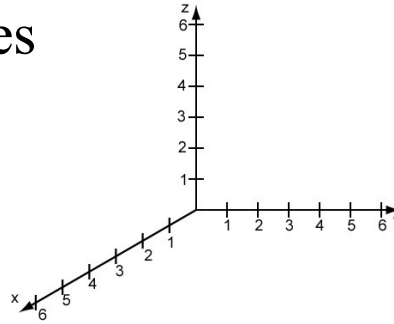
- Crystal planes

Draw the (142) plane.

Invert the Miller indices: $(1, 1/4, 1/2)$

Multiply through by lowest common denominator:

$$4 (1, 1/4, 1/2) = (4, 1, 2).$$



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Examples

- Crystal planes

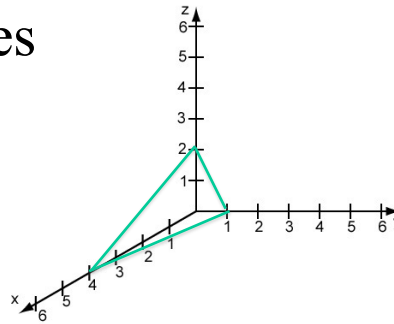
Draw the (142) plane.

Invert the Miller indices: $(1, 1/4, 1/2)$

Multiply through by lowest common denominator:

$$4 (1, 1/4, 1/2) = (4, 1, 2). \text{ These are the x, y, z intercepts.}$$

Draw lines connecting the intercepts.



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Examples

- Crystal planes

For the cubic unit cell, list all of the planes equivalent to (001).

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Examples

- Crystal planes

For the cubic unit cell, list all of the planes equivalent to (001).

(100), ($\bar{1}00$)

(010), ($0\bar{1}0$)

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Examples

- Crystal planes

For the cubic unit cell, list all of the planes equivalent to (001):

(100), ($\bar{1}00$)

(010), ($0\bar{1}0$)

(001), ($00\bar{1}$)

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Examples

- Crystal planes

How do you indicate the full set of planes equivalent to (001) without writing them all out?

(100), ($\bar{1}00$)

(010), ($0\bar{1}0$)

(001), ($00\bar{1}$)

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Examples

- Crystal planes

How do you indicate the full set of planes equivalent to (001) without writing them all out? $\{100\} = \{001\}$

(100), ($\bar{1}$ 00)

(010), (0 $\bar{1}$ 0)

(001), (00 $\bar{1}$)

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Examples

- Crystal planes

Using the correct notation, indicate the set of planes equivalent to (010)?

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Examples

- Crystal planes

Using the correct notation, indicate the set of planes equivalent to (010)? $\{100\}$

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Examples

- Crystal directions

For a cubic crystal, how do you indicate the direction $\hat{x} - \hat{y} + 2\hat{z}$?

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Examples

- Crystal directions

For a cubic crystal, how do you indicate the direction $\hat{x} - \hat{y} + 2\hat{z}$?

$[1\bar{1}2]$

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Examples

- Crystal directions

For a cubic crystal, how do you indicate the direction $-\hat{x} - \hat{y}$?

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Examples

- Crystal directions

For a cubic crystal, how do you indicate the direction $-\hat{x}-\hat{y}$?

$$[\bar{1}\bar{1}0]$$

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Examples

- Crystal directions

For a cubic crystal, how do you indicate the set of equivalent directions to $[\bar{1}\bar{1}0]$?

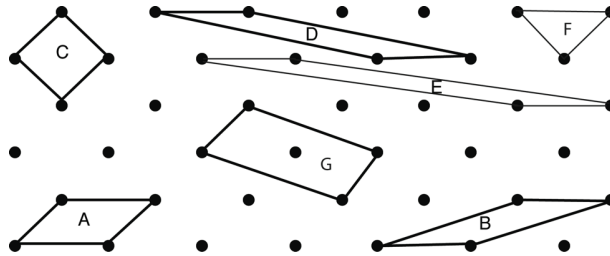
$$\langle 110 \rangle$$

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Examples

- Primitive Cell

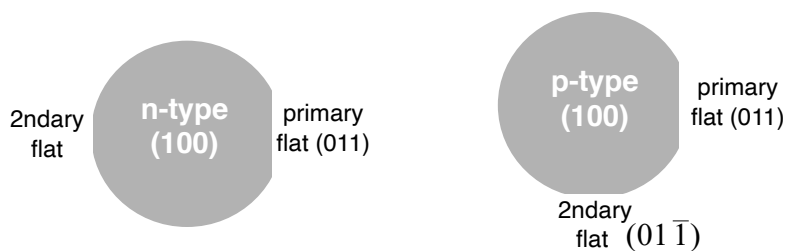
Which ones are valid primitive cells?



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Miller Indices in Everyday Use

- Si Wafers
- Different diameters: 1 in, 2 in, 3 in, 4 in, 6 in, 8 in, 12 in.
- Different orientations:
 - Most common: (100) wafer, i.e. the top surface (plane) is (100).
 - Also order (111) wafers. Top surface is (111).
- Orientation flats on (100) wafers:



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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New Topics

- Energy Bands
 - Conduction and Valence bands
 - Bandgap
- Metals, semiconductors, and insulators
- Electrons and Holes
- Intrinsic and Extrinsic semiconductors
- Substitutional Doping

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Ch. 2. Electrons and Holes

- Start with electrons of isolated Si atom.

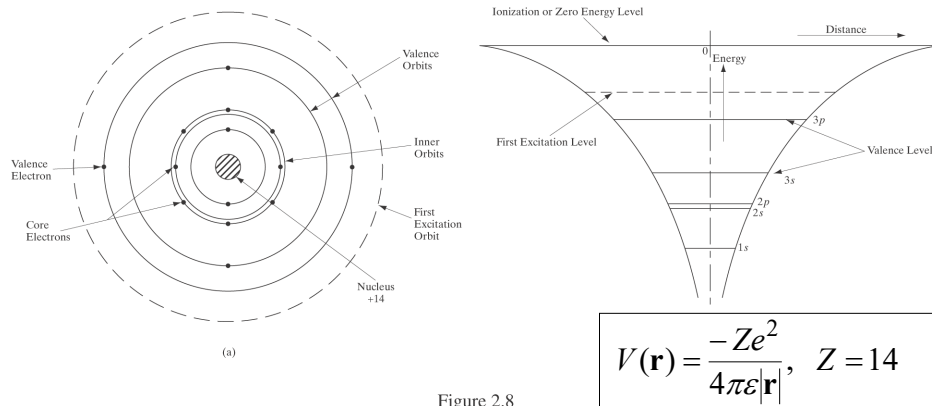


Figure 2.8

Electronic structure and energy levels in a Si atom: (a) The orbital model of a Si atom showing the 10 core electrons ($n = 1$ and 2), and the 4 valence electrons ($n = 3$); (b) energy levels in the coulombic potential of the nucleus are also shown schematically.

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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Valence Electrons

- Electrons that matter to us come from the valence shell of the atom
 - e.g. Si $1s^2 2s^2 2p^6 (3s^2 3p^2)$

core electrons (valence electrons)
 - III-V and II-VI have, on average, 4 electrons in valence shell.
 - Valence electrons participate in the Si – Si covalent bond.

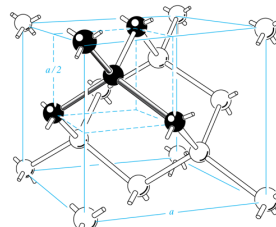
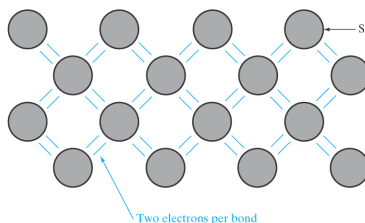
Periodic Table of Elements

1 H																	2 He															
3 Li	4 Be																	19 K	20 Ca													
11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar											72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn

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Si Crystal

- Each Si atom has 4 nearest neighbors.



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

- 2 electrons per bond (represented by double blue lines).
- Each Si atom shares 8 electrons and thus completes its shell as Ar.

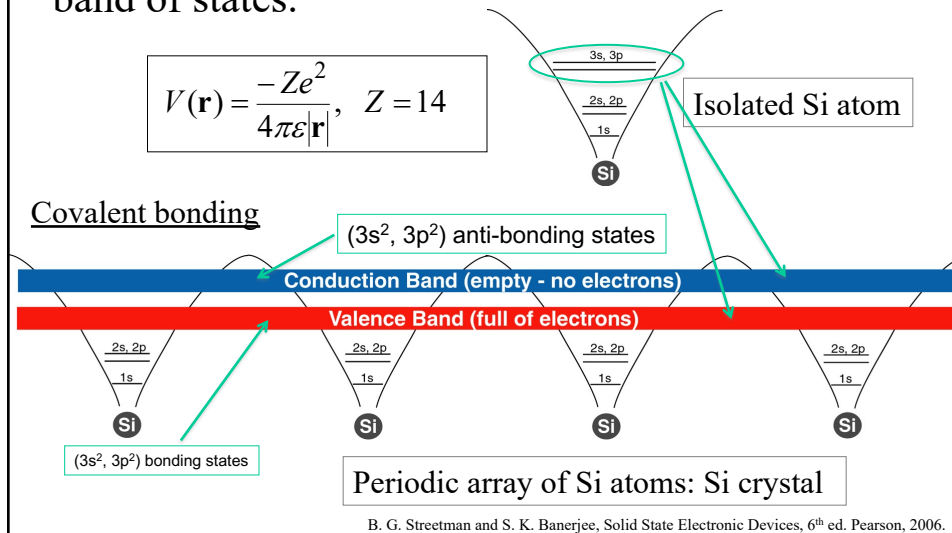
Periodic Table of Elements

1 H	2 He																	18 Ar	19 K	20 Ca											72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	87 Fr	88 Ra	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr
3 Li	4 Be																	19 K	20 Ca											72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	87 Fr	88 Ra	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	
11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar											72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	87 Fr	88 Ra	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr													

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Energy Bands of Semiconductor Crystal

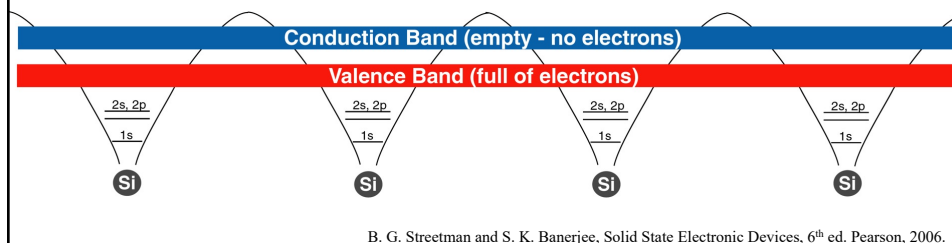
- Discrete atomic valence states ($3s^2 3p^2$) merge into a band of states.



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Energy Bands of Crystal

- N – Si atoms with $4N$ valence electrons (4 foreach Si atom).
- Valence Band
 - Bonding states.
 - Contain the $4N$ electrons of the valence shell.
- Conduction Band
 - Anti-bonding states which are empty of electrons.

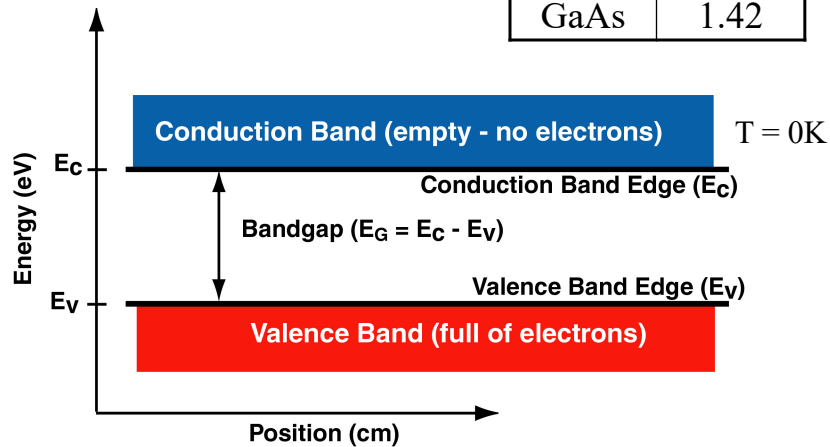


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Conduction and Valence Bands

- E_C Conduction band edge
- E_V Valence band edge
- $E_G = E_C - E_V$ Bandgap

Semi.	E_G (eV)
Si	1.12
Ge	0.66
GaAs	1.42



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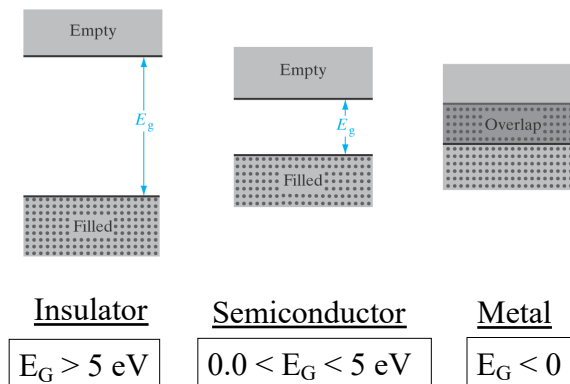
eV

- What is an eV?
 - It is the energy gained (or lost) when one electron passes through a potential drop of 1V.
 - $1 \text{ eV} = 1.602 \times 10^{-19} \text{ C} \times 1 \text{ V} = 1.602 \times 10^{-19} \text{ Joules}$
 - The |charge| of an electron is $q = e = 1.602 \times 10^{-19} \text{ C}$.
 - We will see many quantities like this: $1 \text{ eV} / q = 1 \text{ V}$.
 - $1 \cancel{\text{eV}} / \cancel{e} = 1 \text{ V}$.
 - For example if $kT = 0.026 \cancel{\text{eV}}$, then $kT/\cancel{q} = 0.026 \text{ V}$.

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Insulators, Semiconductors, and Metals

- The bandgap (E_G) distinguishes an insulator from a semiconductor from a metal.

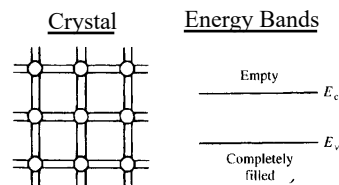


B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

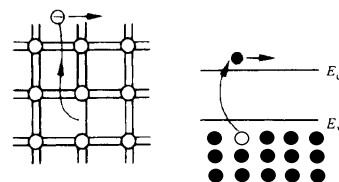
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Electrons

- T = 0K. All of the valence electrons are tied up in the covalent bonds.
 → No free carriers (charge).
 → Acts like insulator.
 → Apply $\mathcal{E}$ -field → NO current.



- Finite T. Thermal energy will break some bonds freeing electrons.
- The energy required is the bandgap.
- On the energy band diagram, the freed electron is promoted from the valence band to the conduction band.
- The freed electron in the conduction band can move around the crystal freely. Becomes a charge carrier.
- Apply $\mathcal{E}$ -field → Current.



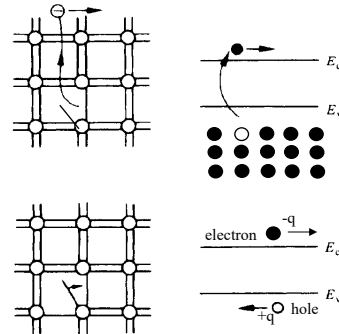
The electron in the conduction band is free to move around.

R. F. Pierret, Semiconductor Device Fundamentals, Addison-Wesley, 1996. Fig. 2.7

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Holes

- The broken bond is also free to move around the lattice. It is a 'missing' bond or 'missing electron.' This missing electron acts like a positive charge and is called a 'hole.'
- On the band diagram, the empty states in the valence band are called holes.
- The holes can freely move, and they act like positively charged electrons.
- The holes will move in response to an electric field and give a current.



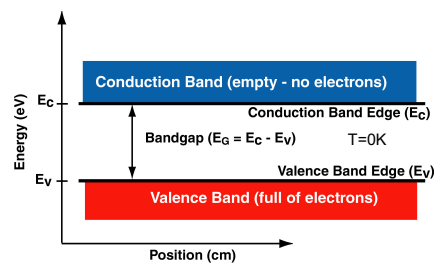
- We are interested in the 'free' carriers which are the
 - Electrons in the conduction band.
 - Holes in the valence band.

R. F. Pierret, Semiconductor Device Fundamentals, Addison-Wesley, 1996, Fig. 2.7

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Review

- The discrete atomic energies merge into Energy bands.
 - Conduction Band** (E_c)
 - Valence Band** (E_v)
 - Band gap** ($E_g = E_c - E_v$)
 - Insulator, semiconductor, and metal** defined by E_g : $E_g > 5\text{eV}$ (insulator), $0.0 < E_g < 5\text{ eV}$ (semiconductor), $E_g < 0$ (metal).
- The bands are plotted as a function of position in the crystal.
- At $T=0\text{K}$, the valence band is full of electrons, and the conduction band is empty.
- At $T=0\text{K}$, an applied electric field gives no current.
 - No current can be carried by an empty band.
 - No current can be carried by a full band.



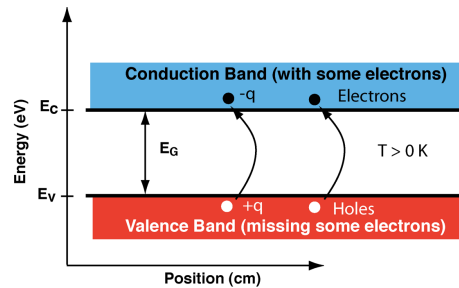
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Review

- Charge Carriers: **Electrons** (-q) and **Holes** (+q)
 - Electrons are the charge carriers in the conduction band
 - Holes are the charge carriers in the valence band.

From a semiconductor device (EE133) perspective:

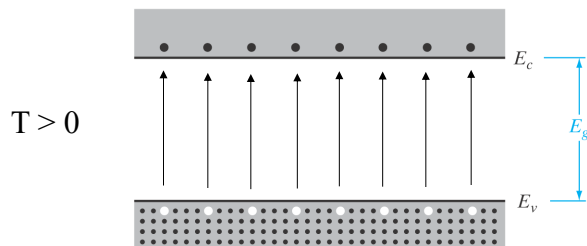
- If there are no electrons in the conduction band, then the electron concentration is 0, i.e. there are no electrons.
- If there are no holes in the valence band, then the hole concentration is 0, i.e. there are no holes.



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Intrinsic Semiconductor

- Pure semiconductor.
- Number of electrons = number of holes → intrinsic



B. G. Streetman and S. K. Banerjee,
Solid State Electronic Devices, 6th
ed. Pearson, 2006.

- Thermal energy breaks bonds in the lattice and creates
 - Electron – Hole pairs
- Every electron that gets promoted from the valence band to the conduction band leaves behind a hole
 - number of electrons = number of holes.
 - net free charge = 0.

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Electron and Hole Density

- Electron density (density of electrons in the conduction band)
 n (1/cm³)
- Hole density (density of holes in the valence band)
 p (1/cm³)
- **Intrinsic semiconductor electron and hole density:**
 $n_i = n = p \leftrightarrow \text{Intrinsic}$
- $n_i = \text{intrinsic carrier concentration.}$ $f(T, E_G)$
- $n_i(T=0) = 0$.

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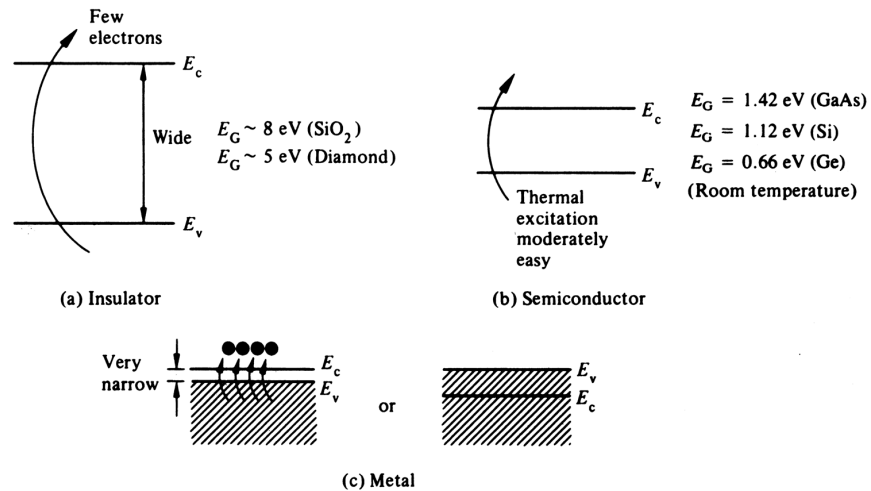
Intrinsic Carrier Concentrations (n_i)

At T=300K:

- Si: $n_i = 1.45 \times 10^{10} / \text{cm}^3$.
 - Ge: $n_i = 2.4 \times 10^{13} / \text{cm}^3$.
 - GaAs: $n_i = 1.8 \times 10^6 / \text{cm}^3$.
- Why do these numbers vary so much?
- For comparison, the density of Si atoms in the Si crystal is $5.0 \times 10^{22} / \text{cm}^3$.
Ge and GaAs: $4.42 \times 10^{22} / \text{cm}^3$.
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Effect of E_G on n_i



R. F. Pierret, Semiconductor Device Fundamentals,
Addison-Wesley, 1996, Fig. 2.7

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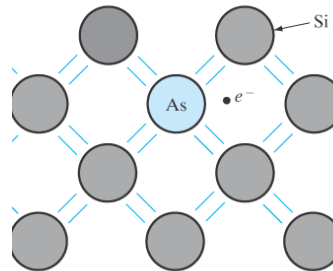
Extrinsic Semiconductor / Doping

- Extrinsic Semiconductor – impurity atoms are introduced to make it mostly n-type or p-type.
- Doping – Process of adding the impurity atoms to increase n or increase p.
- $n \neq n_i$; $p \neq n_i$; $n \neq p$.

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Substitutional Doping of Si

- **n-type dopants** - an atom from column V (P, As, or Sb) replaces a Si atom in the crystal.
- As has 5 valence electrons. It forms 4 bonds with the 4 surrounding Si atoms, and it 'donates' the 5th electron to the crystal. The 5th electron becomes an itinerant electron moving around the conduction band.
- Note that there is no corresponding hole.
- n-type dopants are called **donors**.
- ≈ 1 conduction electron / donor.



As is a donor dopant, an n-type dopant.

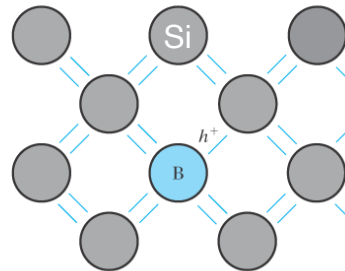
										0
		IIIA	IVA	VA	VIA	VIIA				H
	5	6	7	8	9	10				
	B	C	N	O	F	Ne				
	13	14	15	16	17	18				
	Al	Si	P	S	Cl	Ar				
IB	30	31	32	33	34	35	36			
Zn	Ga	Ge	As	Se	Br	Kr				
	40	50	51	52	53	54				
Cd	In	Sn	Pb	Tl	Po	At	Xe			

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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Substitutional Doping of Si

- **p-type dopants** – an atom from column III (B) replaces a Si atom in the crystal.
- B has 3 valence electrons. It forms 3 bonds with the 4 surrounding Si atoms. The missing bond is free to move around the lattice as a hole in the valence band. Note that there is no corresponding electron in the conduction band.
- p-type dopants are called **acceptors**. They accept an electron from the valence band resulting in a hole.
- ≈ 1 valence hole / acceptor.



B is an acceptor dopant. p-type dopant.

A periodic table of elements with a red circle around Boron (B) in group IIIA, period 2. The table shows elements from Hydrogen (H) to Xenon (Xe) with their atomic numbers and symbols.

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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Substitutional Doping

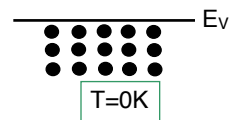
- How do you get the dopants into the lattice?
 - Deposition followed by heat (drive).
 - Ion implantation followed by heat (anneal)

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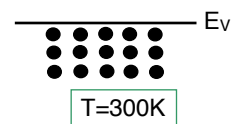
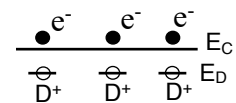
Energy Band View of Dopants

- **Donors; n-type dopants; create electrons in conduction band.**

- The donor level is E_D .
- $E_C - E_D \approx 45 \text{ meV} = 0.045 \text{ eV}$.
- $kT = 26 \text{ meV}$ @ $T=300\text{K}$.
- At $T=0\text{K}$, the 5th electron remains frozen on the dopant atom → no free electrons in conduction band.



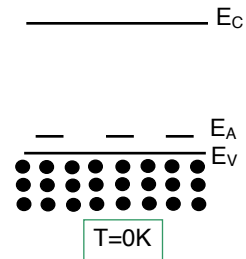
- At room T , 5th electron is thermally excited into the conduction band leaving a + charged donor ion behind.
- The resulting free charge consists of the electrons in the conduction band.
- There are no broken bonds to move around the lattice as holes.



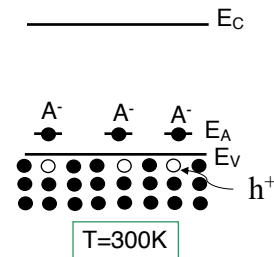
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Energy Band View of Dopants

- **Acceptors; p-type dopants.**
- The acceptor energy level is E_A .
- $E_A - E_V \approx 45 \text{ meV} = 0.045 \text{ eV}$ (Boron).
- $kT = 26 \text{ meV}$ @ $T=300\text{K}$.
- At $T=0\text{K}$, the unsatisfied bond is frozen at the B atom $\rightarrow$ no free holes in valence band.



- At finite T , an electron is thermally excited from the valence band to the acceptor level creating a hole in the valence band and a negatively charged acceptor ion.
- The resulting free positive hole charge is free to move around the valence band.
- The electrons are bound to the acceptor ions and cannot move.



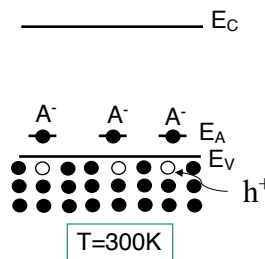
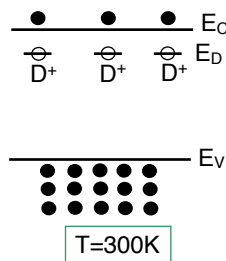
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Dopant Binding Energies (E_B)

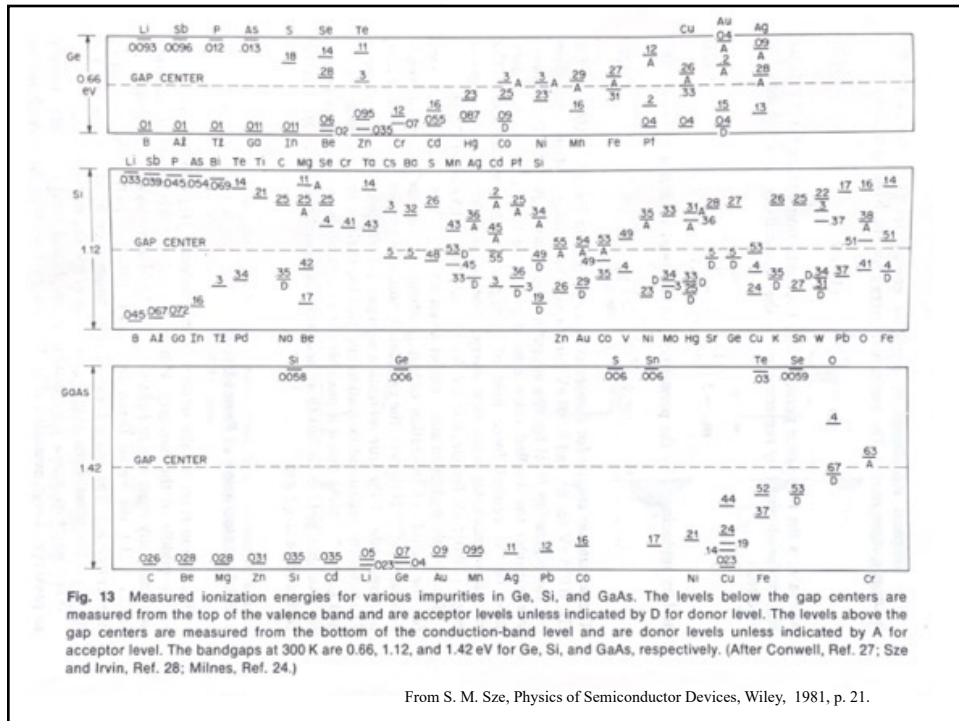
$$E_B = E_C - E_D$$

$$E_B = E_A - E_V$$

Donors	$ E_B \text{ (meV)}$	Acceptors	$ E_B \text{ (meV)}$
Sb	39	B	45
P	45	Al	67
As	54	Ga	72
		In	160

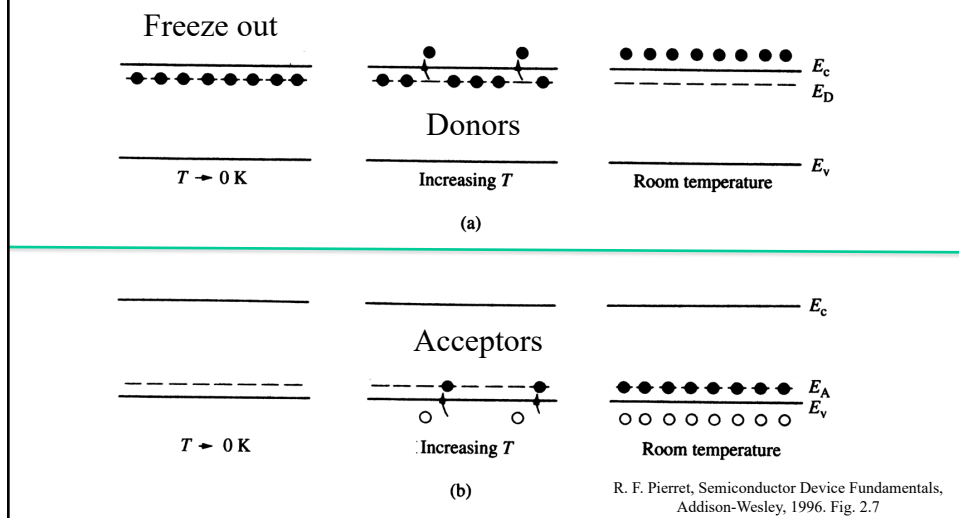


55



56

Effect of Temperature on donors and acceptors: ionization



57

Dopant Concentration

- N_D ($1/\text{cm}^3$) = concentration of donor atoms.
- N_A ($1/\text{cm}^3$) = concentration of acceptor atoms.
- n-type doping
 - $n \approx N_D$.
- p-type doping
 - $p \approx N_A$.

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Review

- The 4 valence electrons of each Si form 4 bonds with 4 nearest neighbors to form the Si crystal.
- At $T=0$, all valence electrons are tied up in the covalent bonds and Si acts like an insulator – there no free charge carriers.
- At $T > 0$, some bonds break creating electrons in the conduction band and holes in the valence band.
- Intrinsic semiconductor: $n = p = n_i$
- Substitutional Doping:
 - An atom from column V with 5 valence electrons substitutes for a Si atom in the lattice. It is a shallow donor. It donates an electron to the conduction band. It is an n-type dopant.
 - An atom from column III with 3 valence electrons substitutes for a Si atom in the lattice. It is a shallow acceptor. It accepts an electron from the filled valence band creating a hole. p-type dopant.
- Extrinsic semiconductor: Doped. $n \neq n_i$; $p \neq n_i$; $n \neq p$.
- Metals, semiconductors, and insulators differ by their bandgap.
- E_C = conduction band edge, E_V = valence band edge, and the bandgap $E_G = E_C - E_V$. The dopant energy levels are denoted E_D and E_A .

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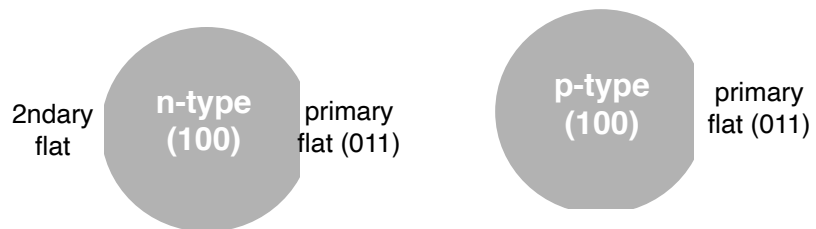
Retrieval Session

- Your supervisor tells you to get a p-type Si wafer
 - How do you identify it?

60

Retrieval Session

- Your supervisor tells you to get a p-type Si wafer
 - How do you identify it?



61

Retrieval Session

- In Si, $n_i = 1.45 \times 10^{10} / \text{cm}^3$ at room temperature.
 - You heat up your wafer and find that in the heated wafer, both n and $p = 1 \times 10^{12} / \text{cm}^3$.
 - Is your Si wafer
 - A. Intrinsic
 - B. Implicit
 - C. Extrinsic
 - D. Explicit
 - E. PG13

62

Retrieval Session

- In Si, $n_i = 1.45 \times 10^{10} / \text{cm}^3$ at room temperature.
 - You heat up your wafer and find that in the heated wafer, both n and $p = 1 \times 10^{12} / \text{cm}^3$.
 - Is your Si wafer
 - A. Intrinsic
 - B. Implicit
 - C. Extrinsic
 - D. Explicit
 - E. PG13
- The semiconductor is INTRINSIC if and only if $n = p$.
- If $n = p$, then it is intrinsic.
- If it is intrinsic, then $n = p$.
- $(n = p) \Leftrightarrow \text{intrinsic}$
- $(n \neq p) \Leftrightarrow \text{extrinsic}$

63

Retrieval Session

- The results of a secondary ion mass spectroscopy (SIMS) test tells you that the P concentration in your Si wafer is $1 \times 10^{18} / \text{cm}^3$ at $T=300\text{K}$. Choose all that apply:
- What is the role of the P in the wafer?
- The wafer is
 - A. n-type
 - B. p-type
 - C. intrinsic
 - D. extrinsic

64

Retrieval Session

- The results of a secondary ion mass spectroscopy (SIMS) test tells you that the P concentration in your Si wafer is $1 \times 10^{18} / \text{cm}^3$ at $T=300\text{K}$. Choose all that apply:
- What is the role of the P in the wafer? **Phosphorous is a dopant. It is in column V and is, therefore, an electron donor.**
- The wafer is
 - A. n-type
 - B. p-type
 - C. intrinsic
 - D. extrinsic

If the Si is doped with an electron donor, then the Si is n-type. If it is n-type, then $n \neq p$, and it is extrinsic.

										2 He
	5 B	6 C	7 N	8 O	9 F	10 Ne				
	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar				
30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr				
48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe				

65

Retrieval Session

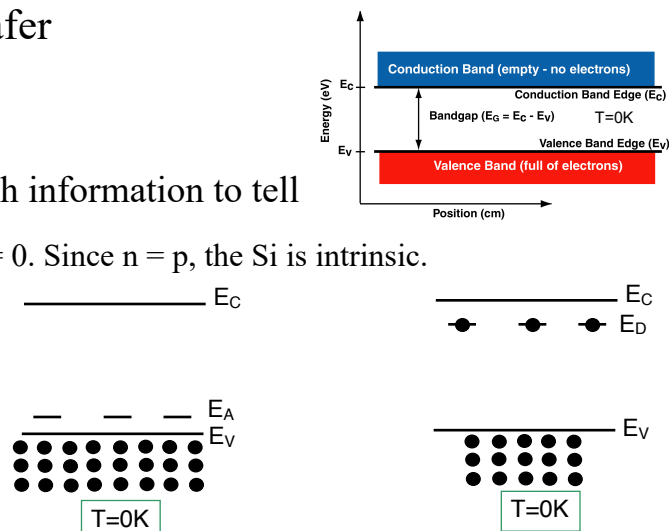
- The temperature of your Si wafer is $T = 0\text{K}$.
- Is your Si wafer
 - A. intrinsic
 - B. extrinsic
 - C. Not enough information to tell

66

Retrieval Session

- The temperature of your Si wafer is $T = 0\text{K}$.
- Is your Si wafer
 - A. intrinsic
 - B. extrinsic
 - C. Not enough information to tell

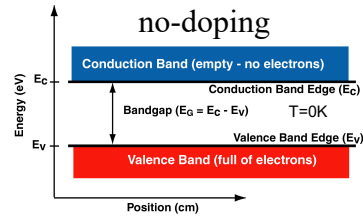
At $T = 0\text{K}$, $n = p = 0$. Since $n = p$, the Si is intrinsic.



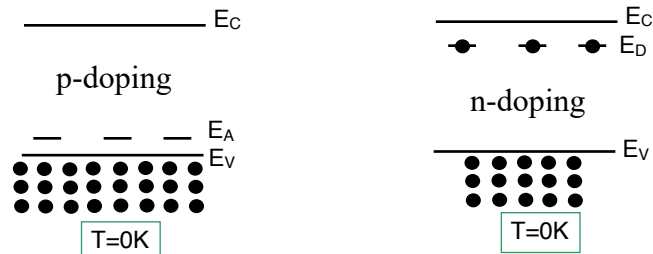
67

Retrieval Session

- The temperature of your Si wafer is $T = 0\text{K}$.
- Is your Si wafer
 - A. intrinsic
 - B. extrinsic
 - C. Not enough information to tell



At $T = 0\text{K}$, $n = p = 0$. Since $n = p$, the Si is intrinsic.



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Retrieval Session

- Given $N_D = 1 \times 10^{18} / \text{cm}^3$ and $T = 300\text{K}$.
- What can you say about your Si wafer?
 - 1.
 - 2.
 - 3.
 - 4.

69

Retrieval Session

- Given $N_D = 1 \times 10^{18} / \text{cm}^3$ and $T = 300\text{K}$.
- What can you say about your Si wafer?
 1. It is n-type.
 2. $n = N_D = 1 \times 10^{18} / \text{cm}^3$.
 3. It is extrinsic.
 4. It is doped with either P, As, or Sb.

70

Retrieval Session

- You want to dope your Si wafer.
 - What are two ways that you can do it?
 - 1.
 - 2.

71

Retrieval Session

- You want to dope your Si wafer.
 - What are two ways that you can do it?
1. Deposition
 2. Ion implantation

72

Retrieval Session

- Both Deposition and Ion Implantation are followed by a heating step. What is the purpose of the heating step in each process?
1. Deposition
 2. Ion implantation

73

Retrieval Session

- Both Deposition and Ion Implantation are followed by a heating step. What is the purpose of the heating step in each process?
 1. Deposition: The heating step (called a 'drive') causes the dopants deposited on the surface to diffuse into the crystal, and also to attain their substitutional positions in the lattice.
 2. Ion implantation: The heating step (called an 'anneal') anneals the damage out of the crystal and allows the dopants to attain their substitutional positions in the lattice.

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Retrieval Session

- What is the room temperature bandgap of
 - Si?
 - Ge?
 - GaAs?

75

Retrieval Session

- What is the room temperature bandgap of
 - Si? 1.12 eV
 - Ge? 0.66 eV
 - GaAs? 1.42 eV
- What is the value of kT at room temperature?

76

Retrieval Session

- What is the room temperature bandgap of
 - Si? 1.12 eV
 - Ge? 0.66 eV
 - GaAs? 1.42 eV
- What is the value of kT at room temperature?
 $kT = 0.026 \text{ eV at } T=300\text{K}.$

77

Retrieval Session

- At $T = 77\text{K}$, what is kT ?

78

Retrieval Session

- At $T = 77\text{K}$, what is kT ?
 - $kT = \underline{0.026} * \underline{77} / 300 = 0.00667 \text{ eV} = \underline{6.67 \text{ meV}}$.
 - $k(T=T_1) = \underline{k(T=300)} * \underline{T_1} / \underline{300}$.

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Retrieval Session

- Approximately, what is the binding energy of B and P in Si?

80

Retrieval Session

- Approximately, what is the binding energy of B and P in Si? 45 meV.

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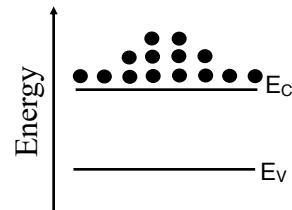
New Topics

- Equilibrium Carrier Concentrations
 - Density of States
 - Fermi function
 - Fermi level
 - Effective Mass
 - Non-degeneracy
 - Effective density of states
 - Intrinsic Carrier concentration
 - Intrinsic Fermi level
 - Intrinsic / Extrinsic

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Carrier Concentration

- Density of states (DOS):
 - Density of available states in the conduction band ($g_c(E)$) and valence band ($g_v(E)$).
 - Function of energy $\sqrt{E}$.



$$g_c(E) = \frac{m_n^* \sqrt{2m_n^*(E - E_C)}}{\pi^2 \hbar^3} \quad (E \geq E_C)$$

Conduction band DOS

$$g_v(E) = \frac{m_p^* \sqrt{2m_p^*(E_V - E)}}{\pi^2 \hbar^3} \quad (E_V \geq E)$$

Valence band DOS

m^* = density of states effective mass

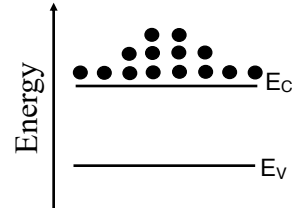
$\hbar$ = Reduced Planck's constant

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Density of States Effective Mass

- m^* :

- Electron (or hole) acts like a ‘free’ electron but with a different mass, m^* .
- Conduction band DOS m^* : m_n^*
- Valence band DOS m^* : m_p^*



Material	DOS m_n^*/m_0	DOS m_p^*/m_0
Si	1.18	0.81
Ge	0.55	0.36
GaAs	0.067	0.52

m_0 = free electron mass.

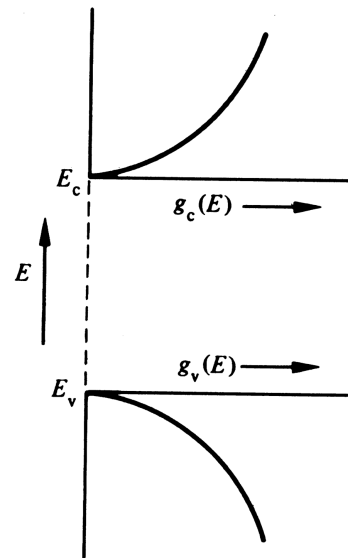
84

Density of States

- Square root of energy.

$$g_c(E) = \frac{m_n^* \sqrt{2m_n^* (E - E_c)}}{\pi^2 \hbar^3}, \quad E \geq E_c$$

$$g_v(E) = \frac{m_p^* \sqrt{2m_p^* (E_v - E)}}{\pi^2 \hbar^3}, \quad E \leq E_v$$



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Fermi Function

- In equilibrium, the probability that a state is occupied with an electron is given by the Fermi function $f(E)$:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

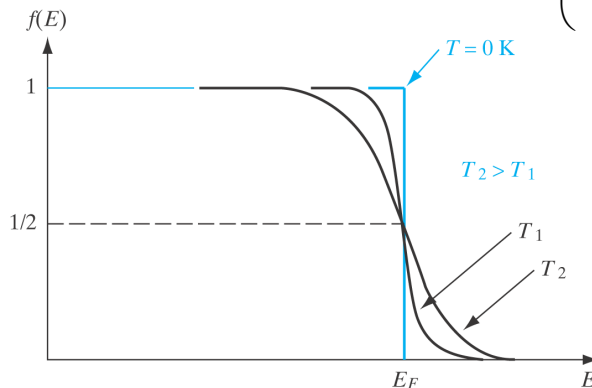
- E_F = Fermi Energy.
- k_B = Boltzmann's constant, 8.617×10^{-5} (eV/K).
- T = temperature (K).
- E = Energy of the electron.
- $f(E \ll E_F) = 1$.
- $f(E = E_F) = 1/2$.
- $f(E \gg E_F) = 0$.

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Fermi Function

- The probability that a state is occupied with an electron

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

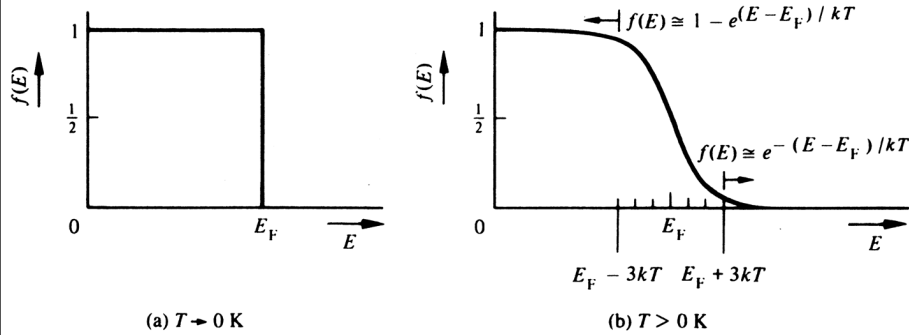


B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

87

Fermi Function

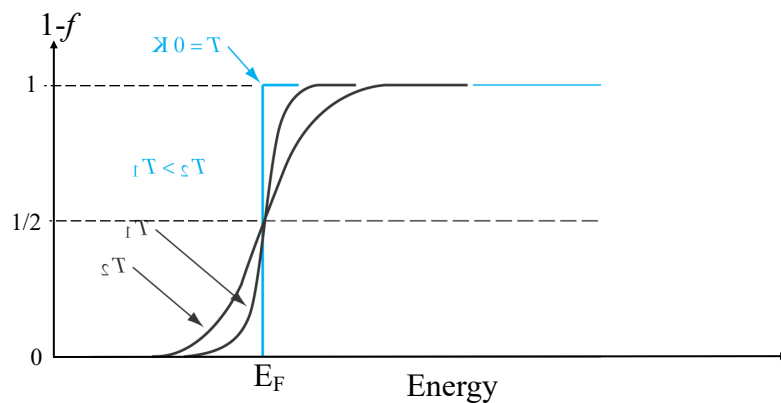
$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$



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Fermi Function

- The probability that a state is occupied with a hole is $[1 - f(E)]$.



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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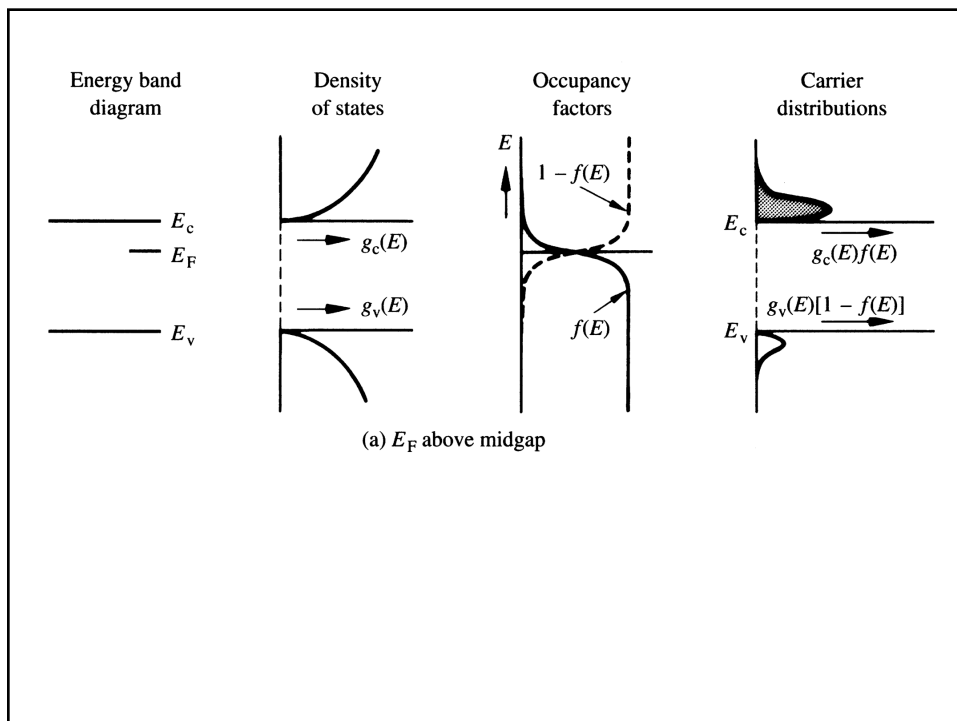
Electron and Hole Density

- Multiply the density of states times the probability that the state is occupied and integrate.

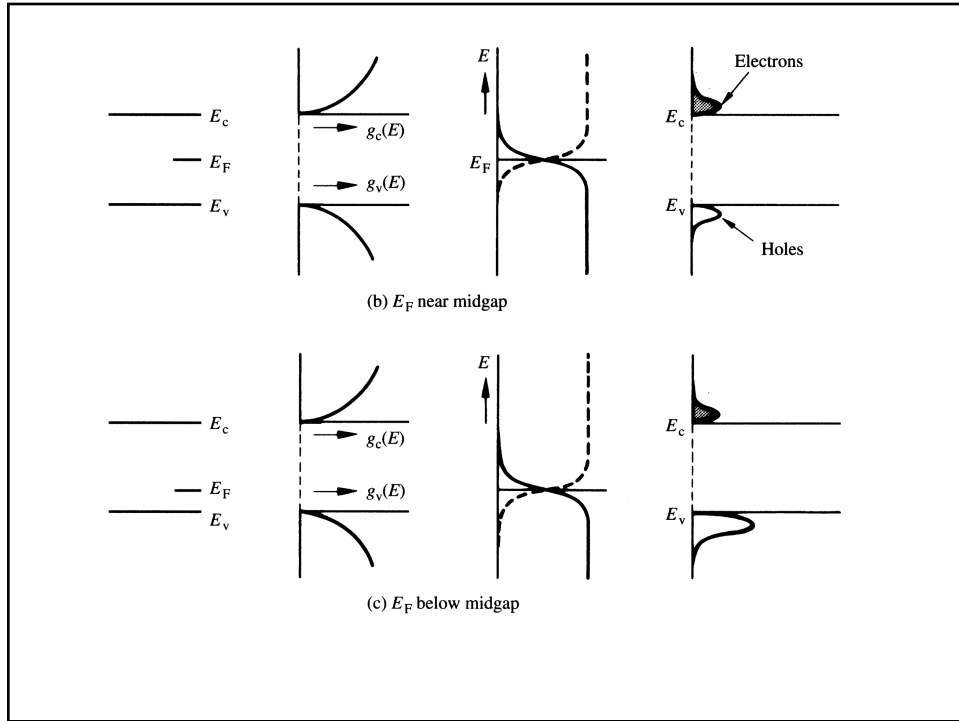
$$n = \int_{\underline{E_C}}^{\infty} g_c(E) f(E) dE$$

$$p = \int_{-\infty}^{E_V} g_v(E) (1 - f(E)) dE$$

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Electron Density

$$n = \int_{E_C}^{\infty} g_c(E) f(E) dE$$

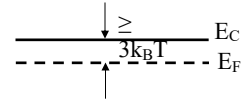
$F_{1/2}$ = Fermi-Dirac integral of order $1/2$.

$$\eta_c = (E_F - E_C) / k_B T$$

$$n = \frac{m_n^* \sqrt{2m_n^*}}{\pi^2 \hbar^3} \int_{E_C}^{\infty} \frac{\sqrt{E - E_C}}{1 + e^{(E - E_F)/k_B T}} dE = N_C \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_c)$$

Non-degenerate

For $E_C - E_F \geq 3 k_B T$, $f(E) \approx e^{-(E - E_F)/k_B T}$.



$$n \approx \frac{m_n^* \sqrt{2m_n^*}}{\pi^2 \hbar^3} \int_{E_C}^{\infty} \sqrt{E - E_C} e^{-(E - E_F)/k_B T} dE$$

E_v

$$n \approx N_C e^{-(E_C - E_F)/k_B T}$$

$$N_C = 2 \left[\frac{m_n^* k_B T}{2\pi \hbar^2} \right]^{3/2}$$

N_C = Effective density of conduction band states.

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Non-Degenerate Electron and Hole Density

Non-degenerate:

- Electrons: $E_C - E_F \geq 3 k_B T$
- Holes: $E_F - E_V \geq 3 k_B T$

$$n \approx N_C e^{-(E_C - E_F)/k_B T}$$

$$N_C = 2 \left[\frac{m_n^* k_B T}{2\pi \hbar^2} \right]^{3/2}$$

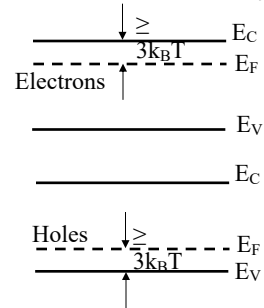
$$p \approx N_V e^{-(E_F - E_V)/k_B T}$$

$$N_V = 2 \left[\frac{m_p^* k_B T}{2\pi \hbar^2} \right]^{3/2}$$

N_C = Effective density of conduction band states.

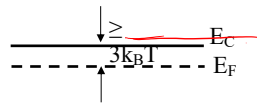
N_V = Effective density of valence band states.

$$N_{C,V} = (2.510 \times 10^{19} / \text{cm}^3) (m_{n,p}^* / m_0)^{3/2} \quad (T = 300\text{K})$$

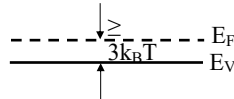


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Non-Degenerate



Non-degenerate iff E_F lies in this region between dashed lines.



Otherwise, it is degenerate.

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Electron and hole density, n and p

- Know this.

$$n = N_C e^{(E_F - E_C)/kT}$$

$$p = N_V e^{(E_V - E_F)/kT}$$

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n and p in Terms of the Intrinsic Carrier Concentration, n_i and Intrinsic Fermi Level, E_i

$$n_i = N_C e^{-(E_C - E_i)/k_B T}$$

$$p_i = N_V e^{-(E_i - E_V)/k_B T} = n_i$$

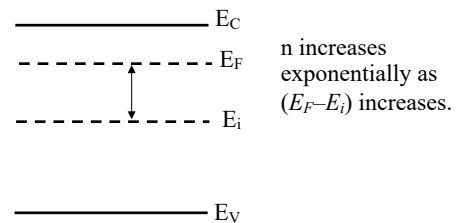
$$N_C = n_i e^{(E_C - E_i)/k_B T}$$

$$n \approx N_C e^{-(E_C - E_F)/k_B T}$$

$$p \approx N_V e^{-(E_F - E_V)/k_B T}$$

$$n = n_i e^{(E_F - E_i)/k_B T}$$

$$p = n_i e^{(E_i - E_F)/k_B T}$$



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n and p

- Know this

$$n = n_i e^{(E_F - E_i)/kT}$$

$$p = n_i e^{(E_i - E_F)/kT}$$

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What is the Value of the intrinsic carrier concentration, n_i ?

$$n_i p_i = N_C e^{-(E_C - E_i)/k_B T} N_V e^{-(E_i - E_V)/k_B T}$$

$$n_i p_i = N_C N_V e^{-E_G/(k_B T)}$$

$$n_i = p_i = \sqrt{N_C N_V} e^{-E_G/(2k_B T)}$$

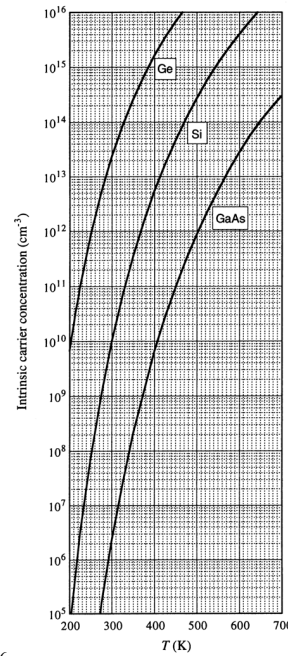
Note the exponential dependence on bandgap and temperature.

Semi.	E_G (eV)	n_i (cm ⁻³) (T=300K)
Si	1.12	1.5e10
Ge	0.66	2.5e13
GaAs	1.42	2e6

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$$n_i(T)$$

- Exponential dependence on $1/T$ and the bandgap, E_G .



Si	
$T(^{\circ}\text{C})$	$n_i (\text{cm}^{-3})$
0	8.86×10^8
5	1.44×10^9
10	2.30×10^9
15	3.62×10^9
20	5.62×10^9
25	8.60×10^9
30	1.30×10^{10}
35	1.93×10^{10}
40	2.85×10^{10}
45	4.15×10^{10}
50	5.97×10^{10}
300 K	1.00×10^{10}

GaAs	
$T(^{\circ}\text{C})$	$n_i (\text{cm}^{-3})$
0	1.02×10^5
5	1.89×10^5
10	3.45×10^5
15	6.15×10^5
20	1.08×10^6
25	1.85×10^6
30	3.13×10^6
35	5.20×10^6
40	8.51×10^6
45	1.37×10^7
50	2.18×10^7
300 K	2.25×10^6

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

Figure 2.20

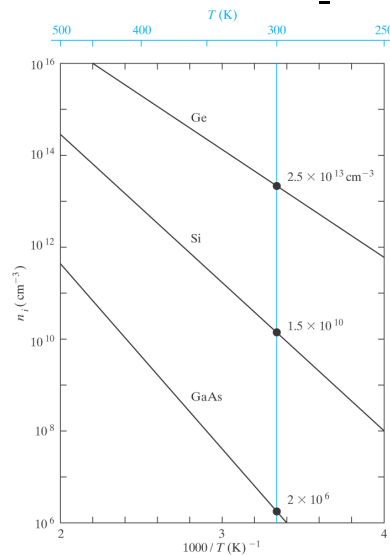
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Temperature Dependence of n_i

Exponential term dominates.

$$n_i = \sqrt{N_C N_V} e^{-E_G/(2k_B T)}$$

$$n_i = 2(m_n^* m_p^*)^{3/4} \left(\frac{2\pi k_B}{h^2} \right)^{3/2} T^{3/2} e^{-E_G/(2k_B T)}$$



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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What is the Value of E_i ?

$$n_i = N_C e^{-(E_C - E_i)/k_B T} = N_V e^{-(E_i - E_V)/k_B T}$$

$$\frac{N_C}{N_V} = e^{-2E_i/(k_B T)} e^{-(E_C + E_V)/k_B T}$$

$$\ln\left(\frac{N_C}{N_V}\right) = [E_C + E_V - 2E_i]/k_B T$$

$$E_i = \frac{E_C + E_V}{2} + \frac{k_B T}{2} \ln\left(\frac{N_V}{N_C}\right)$$

Almost at midgap.

$$E_i = \frac{E_C + E_V}{2} + \frac{3}{4} k_B T \ln\left(\frac{m_p^*}{m_n^*}\right) = \frac{E_C + E_V}{2} - 0.007 \text{ (eV)}$$

Si @ T=300K

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np Product

$$np = n_i^2 e^{(E_F - E_i)/k_B T} e^{(E_i - E_F)/k_B T}$$

$$n = n_i e^{(E_F - E_i)/k_B T}$$

$$np = n_i^2$$

$$p = n_i e^{(E_i - E_F)/k_B T}$$

In equilibrium,

$$np = n_i^2$$

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np product

- Know this
- In equilibrium:

$$np = n_i^2$$

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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Charge Neutrality

- A uniformly doped semiconductor is everywhere charge neutral.

$$q(p - n + N_D^+ - N_A^-) = 0$$

Net charge = 0.

$$p - n + N_D^+ - N_A^- = 0$$

$$p - n + N_D - N_A = 0$$

Assume complete ionization of dopants

$$\frac{n_i^2}{n} - n + N_D - N_A = 0$$

$$np = n_i^2$$

$$n^2 - n(N_D - N_A) - n_i^2 = 0$$

Quadratic equation for n or p.

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

$$p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

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Carrier Concentrations

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

$$p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

- $N_D \gg N_A, n_i$

- $n \approx N_D$
- $p \approx n_i^2 / N_D$

- $N_A \gg N_D, n_i$

- $p \approx N_A$
- $n \approx n_i^2 / N_A$

- Compensation

- $N_A \approx N_D$
- $n \approx p \approx n_i$
- The p-type doping compensates the n-type doping.

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Carrier Concentrations

$$n \approx N_D$$

$$p \approx n_i^2 / N_D$$

$N_D \gg N_A, N_D \gg n_i$
(nondegenerate, total ionization)

$$p \approx N_A$$

$$n \approx n_i^2 / N_A$$

$N_A \gg N_D, N_A \gg n_i$
(nondegenerate, total ionization)

Practical Carrier Conc. Relationships

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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Where is E_F ?

- Normal case of $N_D \gg n_i$, N_A , $n = N_D$.

$$N_D = n_i e^{(E_F - E_i)/k_B T}$$

$$E_F - E_i = k_B T \ln(N_D / n_i)$$

$$n = n_i e^{(E_F - E_i)/k_B T}$$

- $N_A \gg n_i$, N_D , $p = N_A$.

$$N_A = n_i e^{(E_i - E_F)/k_B T}$$

$$E_i - E_F = k_B T \ln(N_A / n_i)$$

$$p = n_i e^{(E_i - E_F)/k_B T}$$

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E_F versus doping

$$E_F - E_i = k_B T \ln(N_D / n_i)$$

$$E_i - E_F = k_B T \ln(N_A / n_i)$$

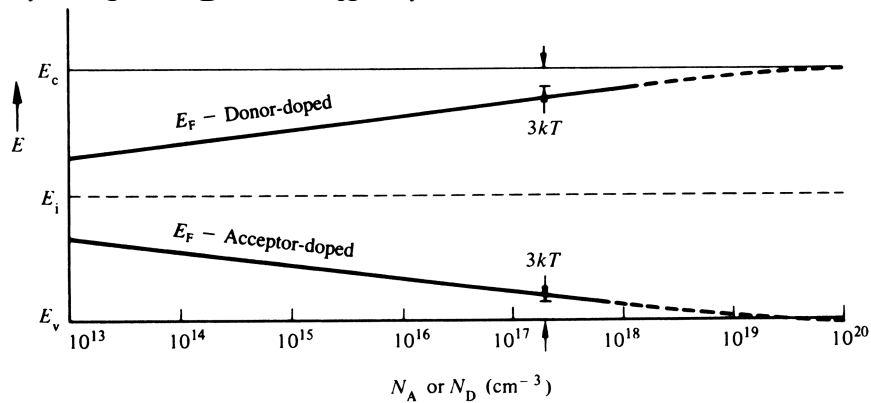
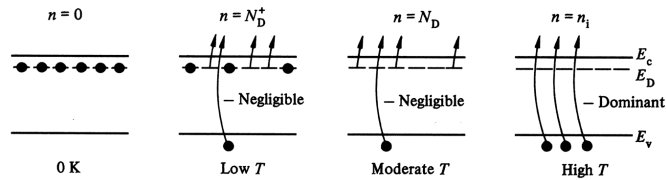
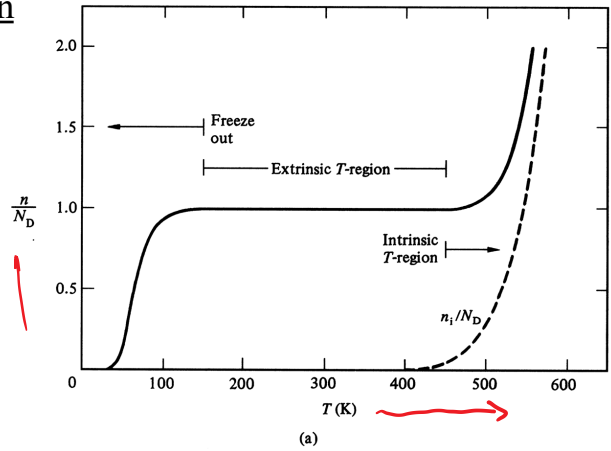


Fig. 2.21 (E_F vs. Dopant Conc.)

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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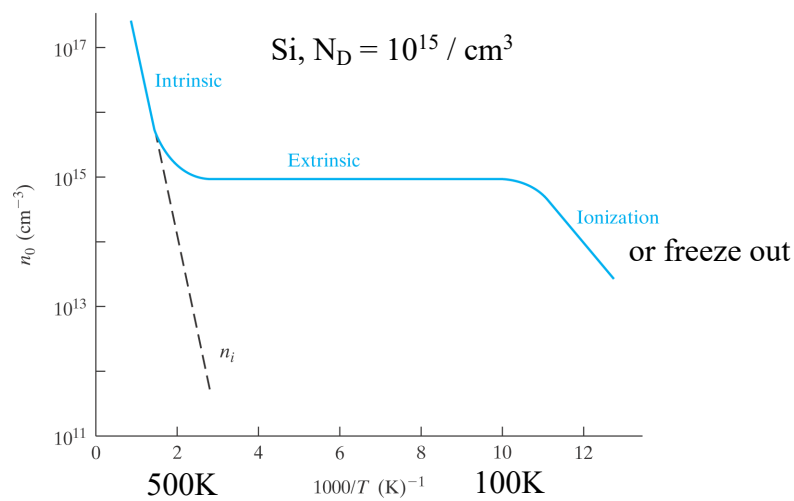
T-Dependence of n



R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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Temperature Dependence of the electron density, n



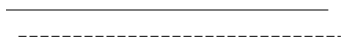
B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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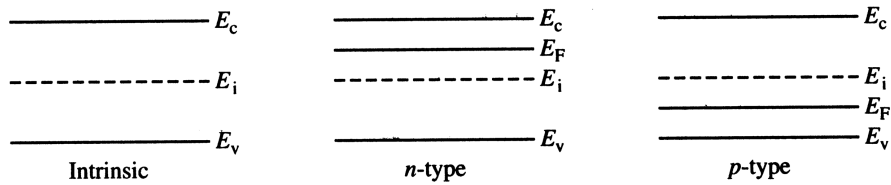
Band diagram

- Draw the band diagram for an n-type semiconductor
- Draw the band diagram for an p-type semiconductor

E_x



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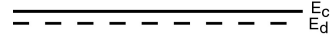


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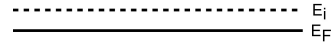
Previous Quiz Problem

Which of the following conditions would give rise to such a band diagram? Explain.

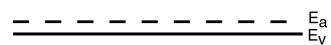
(i) Very high temperature ($kT = EG$)



(ii) High acceptor doping



(iii) Low acceptor doping



(iv) High donor doping

(v) Low donor doping

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Summary of equations.

Table 2.4 Carrier Modeling Equation Summary.		
<i>Density of States and Fermi Function</i>		
$g_c(E) = \frac{m_n^* \sqrt{2m_n^* (E - E_c)}}{\pi^2 \hbar^3}, \quad E \geq E_c$		$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$
$g_v(E) = \frac{m_p^* \sqrt{2m_p^* (E_v - E)}}{\pi^2 \hbar^3}, \quad E \leq E_v$		
<i>Carrier Concentration Relationships</i>		
$n = N_C \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_c)$	$N_C = 2 \left[\frac{m_n^* kT}{2\pi \hbar^2} \right]^{3/2}$	$n = N_C e^{(E_F - E_c)/kT}$
$p = N_V \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_v)$	$N_V = 2 \left[\frac{m_p^* kT}{2\pi \hbar^2} \right]^{3/2}$	$p = N_V e^{(E_v - E_F)/kT}$
<i>n_i, np-Product, and Charge Neutrality</i>		
$n_i = \sqrt{N_C N_V} e^{-E_G/2kT}$	$np = n_i^2$	$p - n + N_D - N_A = 0$
<i>n, p, and Fermi Level Computational Relationships</i>		
$n = \frac{N_D - N_A}{2} + \left[\left(\frac{N_D - N_A}{2} \right)^2 + n_i^2 \right]^{1/2}$		$E_i = \frac{E_c + E_v}{2} + \frac{3}{4} kT \ln \left(\frac{m_p^*}{m_n^*} \right)$
$n \approx N_D$	$N_D \gg N_A, N_D \gg n_i$	$E_F - E_i = kT \ln(n/n_i) = -kT \ln(p/n_i)$
$p \approx n_i^2/N_D$		
$p \approx N_A$	$N_A \gg N_D, N_A \gg n_i$	$E_F - E_i = kT \ln(N_D/n_i) \quad N_D \gg N_A, N_D \gg n_i$
$n \approx n_i^2/N_A$		$E_i - E_F = kT \ln(N_A/n_i) \quad N_A \gg N_D, N_A \gg n_i$

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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